C2.1 Chemical signalling (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **DEFINE** ligand. [C2.1.1] [1]

2 **STATE** one example each of a steroid hormone, a protein hormone and an amine hormone. [C2.1.4] [3]

3 **NAME** the second messenger in the action of adrenaline. [C2.1.10] [1]

4 **STATE** the target cells of progesterone named in the course. [C2.1.13] [1]

5 **NAME** the amino acid phosphorylated by the insulin receptor. [C2.1.11] [1]

6 **DISTINGUISH** between hormones and neurotransmitters in the distance they travel. [C2.1.5] [2]

Section B: Understanding

15 MARKS

7 **EXPLAIN** why steroid hormones have intracellular receptors but insulin has a transmembrane receptor. [C2.1.6] [3]

8 **EXPLAIN** how binding of acetylcholine to its receptor changes the membrane potential. [C2.1.8] [3]

9 **OUTLINE** how a G protein-coupled receptor passes a signal into a cell. [C2.1.9] [3]

10 **EXPLAIN** how insulin increases glucose uptake by a muscle cell. [C2.1.11] [3]

11 **DISTINGUISH** between positive and negative feedback, with an example of each. [C2.1.14] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [C2.1.2, C2.1.14]

Vibrio fischeri was grown in flasks. Cell density and light output per cell were measured every two hours. A second culture was given extra autoinducer from the start.

TIME / H	0	2	4	6	8	10
CELL DENSITY / 10^8 CELLS CM^{-3}	0.1	0.3	1.0	3.0	8.0	10.0
LIGHT PER CELL, NORMAL CULTURE / ARBITRARY UNITS	1	1	2	40	95	100
LIGHT PER CELL, EXTRA AUTOINDUCER ADDED	60	85	95	98	100	100

a DESCRIBE the change in light per cell in the normal culture. [2]

b ESTIMATE the cell density at which light production switches on. [1]

c EXPLAIN the results of the culture given extra autoinducer. [2]

d EXPLAIN why light production rises so steeply once it begins. [2]

e SUGGEST why it is an advantage to the bacteria to make light only at high density. [2]

f STATE one variable that should be kept the same in both cultures. [1]

Section D: Longer answer

6 MARKS

13 COMPARE how adrenaline and oestradiol affect their target cells. [C2.1.10, C2.1.12] [6]